

Requirements Document: AI Video Tagging and Agentic Video Segmentation Solution

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1. Introduction & Objective

GrabTV is seeking a custom, enterprise-grade AI Video Tagging and Agentic Video Segmentation Solution. The objective is to establish a secure, reliable, and scalable video intelligence platform capable of processing multi-minute video streams, detecting objects (specifically focusing on products and people), generating exact bounding boxes, and calculating center coordinates for each detected object.

For the purpose of this project, "video tagging" is defined as agentic video segmentation coupled with the critical secondary step of cataloging identified elements to enable downstream contextual and spatial catalog lookups. The resulting architecture must be modular, highly available, and capable of operating within resource-limited hardware environments on the client consumption side via structured metadata streams.

2. Core Functional Requirements & Capabilities

The Solution Provider/Systems Integrator must design, build, and deploy an end-to-end framework satisfying the following multi-phased lifecycle requirements:

2.1 Environment Assessment & Feasibility

- Conduct a comprehensive tool and technology assessment evaluating standard cloud video indexing APIs against custom computer vision models.
- Define clear operational health indicators, performance metrics, and system thresholds.
- Assess security configurations, role-based access controls (RBAC), authentication mechanisms, and data encryption policies (both in transit and at rest).

2.2 Architecture & High-Level System Design

- **Ingestion Flow:** Design a seamless data pipeline pulling video files from cloud storage backends into an AI analytics engine for real-time or batch segmentation.
- **Object Detection & Frame Extraction:** Implement deep learning models to isolate key video frames, detect humans, and differentiate unique product assets.
- **Bounding Box & Center Calculation Logic:** Define and implement a deterministic algorithm to extract normalized percentage bounding boxes and calculate exact center coordinates (X, Y) for every target object per frame.
- **Custom Vision Training:** Provide an integrated workflow to train, evaluate, and retrain custom vision models using labeled image datasets to support specialized brand recognition or custom logos not covered by stock base models.

2.3 Metadata Stream & API Design

- **JSON Schema:** Define a structured, compact JSON payload schema encapsulating all time-coded video metadata, tags, bounding box coordinates, and classification confidence scores.
- **API Endpoints:** Author secure RESTful API endpoints for orchestrating video ingestion, triggering indexing jobs, and fetching structured metadata payloads.
- **Hardware Optimization:** The output payload must be optimized (and optionally encrypted) for low-overhead consumption by front-end clients, web apps, and resource-limited hardware environments.

2.4 Management, Monitoring & Automation

- Integrate automation logic apps to orchestrate workflows from ingestion to final tagging output.
- Establish localized dashboards to visualize frame mapping, bounding boxes, and product tag associations.
- Implement robust error handling, failure recovery mechanisms for API dropouts, and metadata backup schemes.
- Set up infrastructure monitoring, operational log analytics, alert metrics, and strict cloud cost/budget thresholds.

3. Project Deliverables & Milestones

The Systems Integrator shall deliver the following technical artifacts and validation metrics across the project lifecycle:

Deliverable / Artifact	Phase	Technical Description & Acceptance Criteria
Findings & Recommendations Document	Phase 1: Assess	Comprehensive baseline report outlining tool/technology evaluations, feasibility of base models versus custom vision libraries, initial security configurations, and established operational health indicator thresholds. Deliverable is accepted upon formal joint sign-off.
Solution Architecture & UAT Test Plan	Phase 2: Design	Detailed system topology diagrams, structured JSON schemas for data flow, endpoint definitions for video ingestion APIs, and explicit User Acceptance Testing script logic to evaluate segmentation and bounding box algorithms. Deliverable is accepted upon architecture blueprint verification.

3. Project Deliverables & Milestones, Cont.		
Deliverable / Artifact	Phase	Technical Description & Acceptance Criteria
Deployment Guide & System Runbook	Phase 3: Deploy	An "As-Built" operational execution guide and engineering runbook capturing environment setup parameters, containerized workflow orchestrations, custom training pipeline instructions, dashboard interface connections, and recovery protocols. Deliverable is accepted upon environment provisioning validation.
Completed UAT Validation	Phase 4: Benchmark	Functional verification proving successful parsing of complex multi-minute metadata streams under testing scenarios. Metrics require exact time-coded segmentation bounds, error-free client payload parsing, and a verified performance analytics dashboard. Deliverable is accepted upon pilot deployment confirmation.

3. Project Deliverables & Milestones, Cont.		
Deliverable / Artifact	Phase	Technical Description & Acceptance Criteria
Critical Integration Requirement	Phase 5: Validation	The validation process strictly requires testing and validating agentic video segmentation accuracy by programmatically mapping identified assets to a dynamic "product lookup table." This table must interface directly with a product catalog API chosen at the solution provider or integrator's leisure (e.g., Amazon, Shopify, Kroger, or equivalent vendor ecosystem).

4. Out-of-Scope Services

The following activities are explicitly excluded from this project scope and will be managed independently by GrabTV:

- Troubleshooting existing, internal Line of Business (LOB) 3rd party applications.
- On-premise network infrastructure, Wi-Fi profile customizations, and hardware rack mounting or physical installations.
- Desktop-level user device support and local end-user data backups.
- Providing direct, first-line helpdesk escalation support for GrabTV internal operations.

5. Platform-Agnostic Competitive Matrix

To ensure flexibility and prevent strict vendor lock-in, the solution architecture must remain conceptually generic. The table below outlines the core functional building blocks required for this project alongside direct technological equivalents across major public cloud providers (Microsoft Azure, Amazon Web Services, and Google Cloud Platform).

Functional Component	Microsoft Azure Solution	Amazon Web Services (AWS) Equivalent	Google Cloud Platform (GCP) Equivalent
Core Video Indexing & Insights	Azure AI Video Indexer	Amazon Rekognition Video	GCP Vertex AI Vision / Video Intelligence API
Object Detection & Custom Training	Azure Vision Studio / Custom Vision	Amazon Rekognition Custom Labels	GCP Vertex AI Custom Training (AutoML Vision)
Scalable Object Storage	Azure Storage Accounts (Blob Storage)	Amazon S3 (Simple Storage Service)	Google Cloud Storage (GCS)
Serverless Pipeline Ingestion	Azure Functions	AWS Lambda	GCP Cloud Functions

Functional Component	Microsoft Azure Solution	Amazon Web Services (AWS) Equivalent	Google Cloud Platform (GCP) Equivalent
Application Hosting Environment	Azure App Service	AWS Elastic Beanstalk / Amazon ECS / Fargate	GCP Cloud Run / Google App Engine
Workflow Orchestration & Automation	Azure Logic Apps	AWS Step Functions	GCP Workflows
Operational Monitoring & Log Analytics	Azure Monitor / Log Analytics	Amazon CloudWatch	GCP Cloud Logging / Cloud Monitoring